Guide to Recommended Output Data Formats

AI/ML Hackathon – Ministry of Panchayat Raj (Geospatial Intelligence Challenge)

(Issued by Geo-Intel Lab, IIT Tirupati Navavishkar I-Hub Foundation, with inputs from OGC)

1. Purpose of This Guide

The Geospatial Intelligence Hackathon focuses on AI/ML-driven analysis of **drone imagery and point-cloud data** to support governance-ready applications such as feature extraction, terrain modelling, and drainage network design.

While participants are free to use **any tools, platforms, or workflows**, this guide provides **recommended output data formats** to ensure that hackathon results are:

- Easy to review and validate
- Interoperable across GIS and AI/ML environments
- Reusable by government, academic, and startup ecosystems
- Aligned with widely adopted, open geospatial practices

This guidance **does not change the problem statements, inputs, evaluation criteria, or expected outcomes** of the hackathon. It only addresses **how final outputs are packaged and shared**.

2. Recommended Output Formats

Based on the nature of the problem statements and data involved, the following formats are recommended for final submissions.

2.1 Raster Outputs — Cloud Optimized GeoTIFF (COG)

Applies to:

- Orthophotos
- Digital Terrain Models (DTM)
- Derived raster surfaces (slope, flow accumulation, masks, probability maps, etc.)

Recommended format: Cloud Optimized GeoTIFF (COG)

Why COG:

- Designed for efficient handling of large raster datasets
- Supports fast access, visualization, and processing, especially for large raster datasets and cloud-based workflows
- Widely supported in QGIS, ArcGIS, GDAL, and AI/ML pipelines
- Increasingly used in government and cloud-native geospatial systems

Reference links:

- COG Community Website (tools & ecosystem to generate COG format outputs): https://cogeo.org/
- OGC Cloud Optimized GeoTIFF (COG) Standard: https://www.ogc.org/standards/ogc-cloud-optimized-geotiff/

2.2 Vector Outputs — GeoPackage (GPKG)

Applies to:

- Building footprints (including roof-type attributes)
- Roads
- Drainage networks
- Waterbodies
- Asset locations (DTs, OHTs, wells, etc.)

Recommended format: GeoPackage (GPKG)

Why GeoPackage:

- Single-file, portable container for vector GIS data
- Supports multiple layers, attributes, and CRS in one file
- Easier to manage and review than multi-file formats (e.g., shapefiles)
- Fully supported in QGIS, ArcGIS, and common spatial databases

Reference links:

- GeoPackage Community Website (tools & extensions useful for generating GeoPackage format outputs):
  https://www.geopackage.org/

- OGC GeoPackage Standard:
  https://www.ogc.org/standards/geopackage/

2.3 Point-Cloud / LiDAR Outputs — LAS

Applies to:

- Drone-derived point-cloud data
- Classified point clouds used for DTM generation
- Intermediate or final 3D point-cloud outputs

Recommended format: LAS

Why LAS:

- Widely adopted, open format for LiDAR and point-cloud exchange
- Supported across GIS, surveying, and AI/ML ecosystems through widely used libraries and tools
- Enables consistent sharing of classified and unclassified point data
- Suitable for both research workflows and operational reuse

Reference links:

- LAS format overview:
  https://laspy.readthedocs.io/en/latest/intro.html

- LAS Python library (laspy):
  https://laspy.readthedocs.io/en/latest/index.html

- OGC LAS Standard:
  https://www.ogc.org/standards/las/

3. Application to Hackathon Problem Statements

Problem Statement 1

Feature Extraction from Drone Images (SVAMITVA)

- **Raster inputs and derived outputs**
  → Cloud Optimized GeoTIFF (COG)

- **Extracted features (buildings, roads, waterbodies, assets)**
  → GeoPackage (GPKG)

Problem Statement 2

DTM Creation from Point Clouds and Drainage Network Design

- **Point-cloud data (If providing any processed LiDAR outputs)**
  → LAS
- **DTM and derived raster surfaces**
  → Cloud Optimized GeoTIFF (COG)
- **Drainage networks and GIS-ready design layers**
  → GeoPackage (GPKG)

4. Important Clarifications for Participants

- Participants may use **any software, libraries, or platforms** of their choice
- This guide **does not mandate tools, models, or workflows**
- No additional scoring is tied to format selection
- The intent is to encourage **clean, interoperable, and reusable outputs**
- These formats are already supported by most commonly used GIS and AI/ML tools

5. Why This Matters

By following this guidance, participants help ensure that hackathon outputs:

- Align with **real-world government and academic workflows**
- Can be easily validated, shared, and extended beyond the event
- Introduce practical awareness of **open, vendor-neutral geospatial formats**
- Contribute to a more interoperable and future-ready geospatial ecosystem

This approach reflects IITNiF's broader mission to foster **scalable, standards-aware innovation** while keeping the hackathon inclusive, practical, and outcome-focused.